Creating Accessible Presentations With `ltx-talk`

Timothy Prescott (TeX.SE `teepeemm`)
UNIVERSITY OF NORTH DAKOTA
MATHEMATICS & STATISTICS

July 2026

The Basics

- ```
  \DocumentMetadata{
    lang=en,
    tagging=on,
    pdfstandard=ua-2,
  %  check-tagging-status,
    tagging-setup={
  %    math/alt/use,                % <=> Formulas must have description/alt text
  %    role/new-tag=frametitle/H1, % <=> headings must begin at level 1
      math/setup=mathml-SE
    }
  }
  \documentclass{ltx-talk}
  ```
- LuaLaTeX
- tagged PDF + overlays: `\command<xyz>` → `\command` only on slides xyz
- defined colors: alert, example, structure
- `{frame*}` ⇔ `\verb|...|`, `{verb}`, `{lstlisting}`, etc.
  ```

More Basics — Overlays

- Overlays: \command<xyz> → \command only on slides xyz
- Color commands with overlays: \color, \mathcolor, \textcolor; \alert
- Text commands with overlays: \textbf, \textit, \textmd, \textnormal, \textrm, \textsc, \textsf, \textsl, \texttt, \textup, \emph
- Also: \includegraphics; \label
- Also: \item (and can take an *action spec*)
- Main overlay commands: \onslide, \only, \uncover, \visible, \invisible, \alt, \temporal
- Overlay environments: {onlyenv}, {invisibleenv}, {uncoverenv}, {visibleenv}

A \tableofcontents

Default `tocdepth` is 2 (`\subsection`).

A `\tableofcontents` in a `\section`

Default `tocdepth` is 2 (`\subsection`).

Sectioning

- ~~\part~~, ~~\chapter~~,
  \section, \subsection, \subsubsection,
  ~~\paragraph~~, ~~\subparagraph~~
- (currently?) no visible output
  only affects the \tableofcontents

List Environments

- `{itemize}`, `{enumerate}`, `{description}` as in usual classes
- templates mean we can pass options to the environments, see `texdoc blocks-doc`
- extra (default) key `action-spec`
- `action-spec=<+->` uncover one item at a time
- `action-spec=<+-|alert@+>` and `\alert` the new item

A Strangely Sporadic Glossary

```
\newlength\mylabelwidth
\setlength\mylabelwidth
  {\widthof{tagged PDFs }}\\
\begin{frame}
\frametitle{A Strangely Sporadic Glossary}
 \begin{description}[
  label-width=\mylabelwidth,
  label-align=left,
  left-margin=\mylabelwidth,
  action-spec=<+->, ]
  \item[\LaTeX] \TeX\ and one or two
   additional macros
  \item[\texttt{beamer}] presentation
   class for \LaTeX
  \item[tagged PDFs] Computers can read PDFs
  \item[\texttt{ltx-talk}] tagged
   \texttt{beamer} PDF
 \end{description}
\end{frame}
```

LaTeX TeX and one or two additional macros

beamer presentation class for LaTeX

tagged PDFs Computers can read PDFs

ltx-talk tagged beamer PDF

A Strangely Sporadic Glossary

```
\newlength\mylabelwidth
\setlength\mylabelwidth
  {\widthof{tagged PDFs }}\\
\begin{frame}
\frametitle{A Strangely Sporadic Glossary}
 \begin{description}[
  label-width=\mylabelwidth,
  label-align=left,
  left-margin=\mylabelwidth,
  action-spec=<+->, ]
  \item[\LaTeX] \TeX\ and one or two
   additional macros
  \item[\texttt{beamer}] presentation
   class for \LaTeX
  \item[tagged PDFs] Computers can read PDFs
  \item[\texttt{ltx-talk}] tagged
   \texttt{beamer} PDF
 \end{description}
\end{frame}
```

LaTeX TeX and one or two additional macros

`beamer` presentation class for LaTeX

tagged PDFs Computers can read PDFs

`ltx-talk` tagged `beamer` PDF

A Strangely Sporadic Glossary

```
\newlength\mylabelwidth
\setlength\mylabelwidth
  {\widthof{tagged PDFs }}\\
\begin{frame}
\frametitle{A Strangely Sporadic Glossary}
 \begin{description}[
  label-width=\mylabelwidth,
  label-align=left,
  left-margin=\mylabelwidth,
  action-spec=<+->, ]
  \item[\LaTeX] \TeX\ and one or two
   additional macros
  \item[\texttt{beamer}] presentation
   class for \LaTeX
  \item[tagged PDFs] Computers can read PDFs
  \item[\texttt{ltx-talk}] tagged
   \texttt{beamer} PDF
 \end{description}
\end{frame}
```

LaTeX TeX and one or two additional macros

beamer presentation class for LaTeX

tagged PDFs Computers can read PDFs

ltx-talk tagged beamer PDF

A Strangely Sporadic Glossary

```
\newlength\mylabelwidth
\setlength\mylabelwidth
  {\widthof{tagged PDFs }}\\
\begin{frame}
\frametitle{A Strangely Sporadic Glossary}
 \begin{description}[
  label-width=\mylabelwidth,
  label-align=left,
  left-margin=\mylabelwidth,
  action-spec=<+->, ]
  \item[\LaTeX] \TeX\ and one or two
   additional macros
  \item[\texttt{beamer}] presentation
   class for \LaTeX
  \item[tagged PDFs] Computers can read PDFs
  \item[\texttt{ltx-talk}] tagged
   \texttt{beamer} PDF
 \end{description}
\end{frame}
```

LaTeX TeX and one or two additional macros

`beamer` presentation class for LaTeX

tagged PDFs Computers can read PDFs

`ltx-talk` tagged beamer PDF

Overlays

- Just like in Beamer
- To define an overlay command:

```
\NewDocumentCommand{\mycommand}{ s D<>{1-} ... }{ ...}
% or
\NewDocumentCommand{\mycommand}{ s d<> ... }{ ...}
```

- For example

```
\NewDocumentCommand{\visablealert}{ d<> m }{%
 \visible<#1->{\alert<#1>{#2}}}
```

Matrix Multiplication Example, Step 1

```
\begin{frame}[tag-slides={1-}]
 \frametitle{Matrix Multiplication Example,
  Step \arabic{slide}}
\NewDocumentCommand{\visablealert}{d<>m}{%
 \visible<#1->{\alert<#1>{#2}}}
\newcommand{\colvect}[1]{%
 \begin{pmatrix}#1\end{pmatrix}}
\begin{gather*}
\begin{pmatrix}
\alert<2,3>{1}&\alert<2,3>{2}\\
\alert<4,5>{4}&\alert<4,5>{5}
\end{pmatrix}
\begin{pmatrix}
 \alert<2,4>{-2}&\alert<3,5>{4}\\
 \alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
 {}=\begin{pmatrix}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix}$$

Matrix Multiplication Example, Step 2

```
\NewDocumentCommand{\visablealert}{d<>m}{%
 \visible<#1->{\alert<#1>{#2}}}
\newcommand{\colvect}[1]{%
 \begin{pmatrix}#1\end{pmatrix}}
\begin{gather*}
\begin{pmatrix}
\alert<2,3>{1}&\alert<2,3>{2}\\
\alert<4,5>{4}&\alert<4,5>{5}
\end{pmatrix}
\begin{pmatrix}
 \alert<2,4>{-2}&\alert<3,5>{4}\\
 \alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
 {}=\begin{pmatrix}
  \visablealert<2>{2}&
  \visablealert<3>{-6}\\
  \visablealert<4>{2}&
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & \\ & \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2$$

Matrix Multiplication Example, Step 3

```
\begin{pmatrix}
\alert<2,3>{1}&\alert<2,3>{2}\\
\alert<4,5>{4}&\alert<4,5>{5}
\end{pmatrix}
\begin{pmatrix}
 \alert<2,4>{-2}&\alert<3,5>{4}\\
 \alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
 {}=\begin{pmatrix}
  \visablealert<2>{2}&
  \visablealert<3>{-6}\\
  \visablealert<4>{2}&
  \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1\\2}}%
\visible<2-5>{
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ & \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -5 \end{pmatrix} = -6$$

Matrix Multiplication Example, Step 4

```
\begin{pmatrix}
 \alert<2,4>{-2}&\alert<3,5>{4}\\
 \alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
 {}=\begin{pmatrix}
  \visablealert<2>{2}&
  \visablealert<3>{-6}\\
  \visablealert<4>{2}&
  \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1\\2}}%
\visible<2-5>{
 \alert{
  \only<2,3>{\colvect{1\\2}}
  \only<4,5>{\colvect{4\\5}}
  \bullet
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix} \bullet \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2$$

Matrix Multiplication Example, Step 5

```
\visible<2-6>{
 {}=\begin{pmatrix}
  \visablealert<2>{2}&
  \visablealert<3>{-6}\\
  \visablealert<4>{2}&
  \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1\\2}}%
\visible<2-5>{
 \alert{
  \only<2,3>{\colvect{1\\2}}
  \only<4,5>{\colvect{4\\5}}
  \bullet
  \only<2,4>{\colvect{-2\\2}}
  \only<3,5>{\colvect{4\\-5}}
  =
  \only<2>{2}\only<3>{-6}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ -5 \end{pmatrix} = -9$$

Matrix Multiplication Example, Step 6

```
  \visablealert<3>{-6}\\
  \visablealert<4>{2}&
  \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1\\2}}%
\visible<2-5>{
 \alert{
  \only<2,3>{\colvect{1\\2}}
  \only<4,5>{\colvect{4\\5}}
  \bullet
  \only<2,4>{\colvect{-2\\2}}
  \only<3,5>{\colvect{4\\-5}}
  =
  \only<2>{2}\only<3>{-6}
  \only<4>{2}\only<5>{-9}}}
\end{gather*}
\end{frame}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & -9 \end{pmatrix}$$

Which slides to use?

- Which slides should a screen reader use?
- Default is the last slide: `tag-slides=n`
- Previous example: `tag-slides={1-}`

- Which slides should `\documentclass[mode=handout]{ltx-talk}` use?
- Default is to ignore all overlays
- Fails in previous example:

$$\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}4\\5\end{pmatrix}\cdot\begin{pmatrix}-2\\2\end{pmatrix}\begin{pmatrix}4\\-5\end{pmatrix}=2-62-9$$

Adjusting the Header

`\EditInstance{header}{std}{<key=value>}` or
`\EditInstance{frametitle}{header}{font=<value>}`

Key	Default Value
background-color	(none)
color	structure
font	`\normalfont`
height	topmargin (10 mm) + headsep (2 mm)
left-hspace	leftmargin (10 mm)
right-hspace	rightmargin (10 mm)
print-frame-title	true

A Logo in the Header $\longrightarrow$ $\longrightarrow$

```
\NewCommandCopy\oldframetitle\frametitle
\renewcommand{\frametitle}[1]{%
  \oldframetitle{#1\hfill
    \includegraphics[height=2ex,artifact]{logo-file}%
  }%
}
```

Adjusting the Footer

`\EditInstance{footer}{std}{<key=value>}`

Key	Default Value
background-color	(none)
color	black
element-order	(empty)
font	`\tiny`
left-hspace	leftmargin (10 mm)
right-hspace	rightmargin (10 mm)
separator	`\hfil`

Available footer elements:
author, date, institute, title, subtitle, framenumber, totalframes (=`\RefProperty{lastpage}{totalframes}`)
For example:

```
\EditInstance{footer}{std}{element-order={date,author,title,framenumber}}
```

Adjusting the Title Page

```
\maketitle[
  element-order          = {
    title, subtitle, author, institute, date
  },
  framestyle             = talk,
  horizontal-alignment = center,
  vertical-alignment   = center
]
```

framestyle = pagestyle

other options are plain & wallpaper

Creating a Custom Title Element

```
\DeclareInstance{titlepage-element}{myelement}{talk}{<key=value>}
% and then
\makeatletter
\newcommand\@myelement{...}
\makeatother
% or
\ExpandArgs{c}\newcommand{@myelement}{...}
```

Creating a Custom Footer Element

```
\DeclareInstance{footer-element}{myelement}{talk}{...}
% and then
\makeatletter
\newcommand\@myelement{...}
% or
\newcommand\@shortmyelement{...}
\makeatother
% or
\ExpandArgs{c}\newcommand{@myelement}{...}
% or
\ExpandArgs{c}\newcommand{@shortmyelement}{...}
```

(`\@shortmyelement` will be preferred if available)

More Customizations

```
\EditInstance{floatenv}{std}{horizontal-alignment = <alignment>}
% left (default), center, or right

\EditInstance{hidden}{std}{opacity = <value>}
% 0 <= value <= 1 for how opaque hidden things should be
```

Future Work

- better tagging
- better heading structure?